Mainendra Patel

Passionate Developer | JavaScript & TypeScript | Video Applications



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Passionate developer with deep expertise in JavaScript, TypeScript and modern web technologies. Skilled in creating scalable, user-friendly, and accessible applications across platforms such as desktop, mobile, browsers, smart TVs, and cloud environments. Strong problem-solving, debugging, and collaboration skills with a focus on quality, performance, and user experience.

PROJECT AND TOOLING

Video Navigator - Cross-Platform streaming application

Navigator application to consume video on interactive TVs like Samsung, LG, Vizio, GoogleTV, game consoles like Xbox and streaming devices like Chromecast, FireTV, etc. Users can watch Live TV, OnDemand Movies, record airing programs, and purchase Movies / TV Shows. Supports playback types such as Free OnDemand, Transactional OnDemand, Cloud DVR and Live TV. Player supports Dynamic Ad Insertion and trick modes (play/pause, seek, restart), plus In-App Buy flow, cloud/set-top-box recording, and parental controls. Uses the Speech Synthesis web API for accessibility, web app supports NVDA and JAWS screen readers on Windows and VoiceOver on Mac.

Responsible for architecting, designing, and development with accessibility support across platforms. Collaborated with clients to understand business requirements and maintained design documentation. Provided development estimates, prioritized and assigned tasks, and implemented best practices for coding standards, CI/CD, and deployment. Conducted code reviews and mentored junior developers to ensure code quality and team growth.

Developer Toolkit - Internal Developer Tools Suite (Desktop App + Sidecars)

Designed and built a cross-platform desktop toolkit to streamline development, debugging, and QA workflows for a video SDK team. Built as a Tauri 2 desktop app (SvelteKit 2 + Svelte 5 Runes + TypeScript + Tailwind CSS v4) with a minimal Rust backend, bundled with two standalone Bun-compiled sidecar binaries: a local proxy server and a WebSocket server. The suite provides tabbed tooling for proxy control, WebSocket control, and test automation. Shipped two build variants from a single codebase - a full edition and a lightweight proxy-only edition - driven by environment flags and separate build configs. Implemented a custom DMG-based auto-update mechanism backed by a private package registry, versioned release pipelines (patch/minor/major with synchronized versioning), and clean process lifecycle management. State persisted via the Tauri store plugin and localStorage with no external state library - all reactivity handled through Svelte 5 runes.

SKILLS

- **Languages:** JavaScript, TypeScript, HTML5, CSS3, Rust*, Python*, Ruby*, Java*
- **Frameworks / Libraries:** React, Angular, BackboneJS, SolidJS*, Svelte 5, LightningJS, Node.js, Bun, Tauri, REST APIs, GraphQL*
- **Tools / Platforms:** Vite, Webpack, Gulp, Git, GitHub, GitLab, Jira, Kibana, Tableau, Docker, Tailwind CSS
- **Accessibility:** WCAG, NVDA, JAWS, Speech Synthesis
- **Practices:** Agile, Scrum, CI/CD, TDD/BDD, Event-driven & Async Programming, Debugging, Problem Solving

PROFESSIONAL EXPERIENCE

Technical Lead / Functional Specialist - Cognizant | Present
Technical Lead - itaas - A Cognizant Company
Associate Software Engineer - Spartan Labs Pvt. Ltd. | Earlier Experience